

Monopile Installation Record HB-A13

Harrow Bay Offshore Wind · Foundations

Elena Márquez · 15 September 2026 · HB-CON-IR-A13

Position	Date	Vessel	Contract
HB-A13	11 Sep 2026	MV Sea Atlas	WE-CON-2025-040 / -027

1. Position and pile data

Item	Design	As-installed
Coordinates (WGS84)	53.93720 N, 1.23260 E	within 26 cm of target
Water depth (m LAT)	35.1	34.8
Pile length (m)	79.1	79.1
Pile mass (t)	1,396	–
Target penetration (m)	36.6	36.9
Verticality	≤ 0.25°	0.12°

2. Driving record

Item	Value
Start of soft start	11:02
Driving duration	4 h 56 min
Total blow count	5,884
Maximum hammer energy	2,758 kJ
Self-weight penetration	3.2 m
Interruptions	None

3. Noise and marine mammal mitigation

Item	Value
Bubble curtain	Double big bubble curtain, both rings operational
Peak SEL at 750 m	158.7 dB (limit 160 dB)
Pre-piling search (MMO + PAM)	30 min, marine mammal detected – start delayed 20 min
Acoustic deterrent device	Activated 30 min before soft start

4. Transition piece

Item	Value
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TP installed	14 Sep 2026
Flange flatness after installation	1.0 mm (limit 2.0 mm)
Bolt tensioning	148 × M72 – scheduled with next campaign
TP level	+0.03° (limit 0.25°)

5. Non-conformances

No non-conformances raised.

6. Sign-off

Role	Name	Date
Offshore Installation Supervisor	Atlas Marine Contractors	12 Sep 2026
Client Representative	Felix Brandt	12 Sep 2026
Reviewed	Elena Márquez	13 Sep 2026